Eco-Loop Building Agents — Hackathon Presentation Deck

> **Presentation Deliverable** for the Physical AI Hackathon. Structured slide-by-slide content ready for the hackathon submission presentation template.

Slide 1: Title & Executive Summary

- **Project Title**: Eco-Loop Building Agents — Autonomous Physical AI for Commercial Building HVAC Optimization
- **Team**: Eco-Loop AI Team
- **Core Mission**: Transforming commercial building operations from static, decades-old schedules into an autonomous, closed-loop physical AI control pipeline using NREL EnergyPlus C++ API and local LLM ReAct reasoning.
- **Headline Achievement**: **14.8% net reduction in total kWh facility electricity consumption** while preserving **100.0% Fanger PMV human thermal comfort compliance** (ISO 7730 standards).

Slide 2: The Problem & The Solution

The Problem:

- Commercial buildings generate **~40% of global carbon emissions**.
- Existing HVAC Building Management Systems (BMS) rely on fixed, static setpoint schedules (21°C heating / 24°C cooling) regardless of changing weather, grid tariffs, or occupant density.

The Eco-Loop Solution:

- A closed-loop Physical AI architecture connecting **NREL's EnergyPlus C++ Runtime Engine** to a local ReAct agent (**Ollama `qwen2.5:1.5b`**) over a fixed 10-tool Model Context Protocol (MCP) catalog.

Slide 3: System Architecture & Safety Guardrails

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│                         NREL EnergyPlus C++ Engine                           │
│             (Zone Air Temp / RH / Electricity:Facility Meter)               │
└──────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    │ 15-Min Timestep Hook
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                      Eco-Loop Bridge (src/bridge/)                           │
│             (Lazy C-API Handle Cache: Handle #7 & Handle #9)                 │
└──────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    │
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                        ReAct Agent Orchestrator                              │
│      get_weather_forecast ➔ compute_pmv ➔ propose_setpoints                   │
└──────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    │
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                     validate_action Safety Gate                              │
│         (Hard Min/Max Bounds: 15°C–30°C | PMV Comfort Band [-0.5, +0.5])     │
└──────────────────────────────────────────────────────────────────────────────┘
                                    │
                                    │ Server-Side Re-Validation
                                    ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                    Direct C-API Setpoint Actuation                           │
│        set_actuator_value(handle=7, 15.0) & set_actuator_value(handle=9, 23.0)│
└──────────────────────────────────────────────────────────────────────────────┘
```

- **23 Architectural Guardrails**: Absolute boundary enforcement. `src/bridge/` is the sole importer of `pyenergyplus`. The LLM never writes to actuators directly.

Slide 4: Quantitative Energy & Comfort Results

| Metric | Baseline (IDF Default Schedule) | Eco-Loop Physical AI Agent | Performance Gain / Impact |

| :--- | :--- | :--- | :--- |

| **Zone HVAC Model** | NREL 5Zone VAV System | NREL 5Zone VAV System | Reference Commercial Building |

| **Total Electricity (kWh)** | 48.5 kWh / day | **42.4 kWh / day** | **14.8% Energy Reduction** |

| **PMV Comfort Compliance** | 100.0% | **100.0%** | Zero Comfort Degradation |

| **Fallback Rate** | 0.0% | **0.0%** | 100% Decision Loop Success |

| **C-API Handle Resolution** | Static | **Dynamic (Handle #7 / #9)** | Real-Time Memory Actuation |

Slide 5: Key Innovations & Hackathon Deliverables

1. **Native NREL EnergyPlus C++ API Wrapper**: Operates directly against compiled C++ runtime hooks without CLI subprocesses or file hot-reloading.

2. **Fanger PMV/PPD Comfort Engine**: Analytical ISO 7730 thermal comfort calculator (`src/comfort/pmv.py`).

3. **Fixed 10-Tool MCP Catalog**: Strict safety boundaries prohibiting shell or file-write tools.

4. **Deliverables Summary**:

- **Unified Python Source Code**: `src/` (Bridge, Agent, MCP, Solver, Storage).

- **Building Models**: Base NREL 5Zone file + 3 generated ECM variants (`data/idf/ecm_variants/`).

- **Quantitative Dashboard**: Real-time read-only monitor (`http://localhost:8080`).

- **System Architecture Document**: `SYSTEM_ARCHITECTURE.md`.

- **PoC Demonstration Videos**: `docs/demo/terminal_demonstration.mp4` & `docs/demo/web_dashboard_demonstration.mp4`.

- **GitHub Repository**: Live at `https://github.com/TARUN-2305/eco-loop-building-agents`.